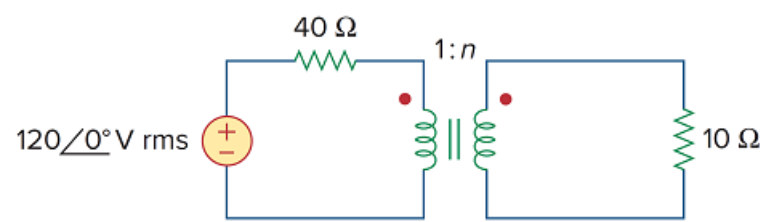


Problem

For the circuit in Fig. 13.117, determine the turns ratio n that will cause maximum average power transfer to the load. Calculate that maximum average power.

Figure 13.117



Step-by-step solution

Step 1 of 2

Refer to figure 13.117 given in the textbook.

Compare the circuit shown in figure 13.117 with the circuit shown in figure 13.33 given in the textbook.

For the maximum power transfer from source to the load, refer to equation 13.60 given in the textbook.

$$Z_{in} = \frac{Z_L}{n^2}$$

Substitute 40 for Z_{in} , and 10 for Z_L .

$$40 = \frac{10}{n^2}$$

$$n^2 = \frac{1}{4}$$

$$n = \frac{1}{2}$$

$$n = 0.5$$

Therefore, the turns ratio n that will cause maximum average power transfer to the load is

$$\boxed{0.5}.$$

Step 2 of 2

Now, the impedance transfers from source to load for maximum power transfer is,

$$Z_L = 40 \, \Omega$$

Calculate the maximum average power.

$$P = \frac{V_{rms}^2}{4Z_L}$$

Substitute 120 for V_{rms} , and 40 for Z_L .

$$\begin{aligned} P &= \frac{(120)^2}{4(40)} \\ &= \frac{14400}{160} \\ &= 90 \, \text{W} \end{aligned}$$

Therefore, the maximum average power is 90 W.